REMZI EFE DEMIR

Mechatronics Engineering Student | Embedded Systems, Power Electronics & UAVs

Electronics and UAV team lead: captained a 17-person swarm UAV team and now leads two autonomous drone projects, one approved by HAVELSAN and AFAD.

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HAVELSAN & AFAD	TÜBİTAK 2209-B	MÜBAK '25 & '26
Reconnaissance drone approved	Research project accepted, ongoing	2 conference talks, 1 co-authored paper

EDUCATION

Atılım University - Ankara, Türkiye · **B.Sc. Mechatronics Engineering** *Sep 2023 - Expected Jun 2027*

- Relevant coursework: Electronics, Microprocessors (PIC18 Assembly), Microcontroller Programming, Manufacturing Processes, Differential Equations

SKILLS

Electronics & Embedded: PCB design (Altium Designer), power electronics / SMPS (flyback), LTspice, BMS design, STM32, ESP32, Arduino, Raspberry Pi, NVIDIA Jetson, Embedded C/C++, I2C / SPI / UART, RF communication
Autonomy & Software: Python, MATLAB, ROS, PID control, Kalman filter, reinforcement learning, computer vision (OpenCV, YOLOv5), swarm algorithms, sim-to-real, Linux
Mechanical & Process: SolidWorks, Siemens NX, 3D printing, MIL-STD-810, MIL-STD-498, SOP, Lastenheft (requirements specification), IFS ERP
Languages: Turkish (Native) • English (B2, Upper Intermediate) • Danish (Basic)

EXPERIENCE

Neutronix Technologies - Remote · **Voluntary Intern, Power Electronics** *Aug 2026 - Present*

- Designed a 40 W flyback switched-mode power supply end to end: converter design and LTspice simulation, schematic and PCB layout in Altium Designer; prototype bench testing is the next stage.

MZD Elektrik Elektronik A.Ş. · **Mechatronics Engineering Intern** *Jul 2026 - Aug 2026*

- Rotated through 4 departments of a defense cable-harness manufacturer and ROKETSAN strategic partner over 20 working days.
- Followed production planning end to end in IFS ERP, from purchasing work order to shipment, and studied shop-floor SOPs.

QuadRobotics R&D and Software · **Voluntary Intern** *Jan 2025 - Mar 2025*

- Supported payload integration on the ROKETSAN KOZ quadruped platform with SolidWorks CAD; checked design compliance with MIL-STD-810 and MIL-STD-498.
- Contributed to payload hardware integration on robot dog platforms, NACA-based airfoil design and testing of autonomous robotic systems.

PROJECTS

High-Autonomy Reconnaissance Drone · **Project Lead • 4-person team** *2026 - Present*

- Developing a high-autonomy reconnaissance drone; the project is approved by HAVELSAN and AFAD. Technical details are confidential.

Autonomous Tracking & Moving-Platform Landing UAV · **Project Leader • 6-person team** *2026 - Present*

- Leading a 6-person team developing a UAV that autonomously tracks a target and takes off from and lands on a moving platform; completion targeted for 2027.

Swarm UAV for GPS-Denied Environments · **Team Captain • 17-person team** *Nov 2024 - Aug 2026*

- Founded a self-funded 17-person team to build a low-cost indoor drone swarm; reached TEKNOFEST Swarm UAV and passed the preliminary design report and video stages; incubated at Atılım University's pre-incubation center.
- Built the flight stack in-house: 3D-printed SolidWorks airframe, Altium flight-controller carrier PCB with power distribution, CRC-checked RF remote and own PID flight controller on ESP32.
- Accepted for TÜBİTAK 2209-B (2025 - present): concept adapted to a single drone with autonomous takeoff and landing for environmental data collection at sea.

Autonomous Underwater Vehicle (AUV) · **Electronics Team Leader** *Sep 2024 - Aug 2026*

- Designed power electronics, custom circuitry and battery system; led PCB design, sensor integration and communications; Kalman-filtered IMU with PID orientation control; anomaly detection with OpenCV and YOLOv5.
- Promoted from team member to electronics team leader within 3 months; co-authored the MÜBAK'25 paper and presented the vehicle at MÜBAK'26.

Unmanned Surface Vehicle (USV) · **Electronics Team Leader** *2024 - 2025*

- Designed the electronic architecture and schematics in Altium Designer and the RF link to the kamikaze drones carried on board; presented the project at MÜBAK'25.

Wearable Health Tracking Device · **Embedded Software** *2024 - 2025*

- Implemented I2C, SPI and UART sensor data acquisition for a dysautonomia-monitoring graduation project that won 1st prize at Ankara University.